Experiment 7:. Random Sampling and Sampling Distribution

Aim :

To explore random sampling from a population and understand the concept of sampling distribution using Python in Jupyter Notebook.

Program Code :

```
import numpy as np

import matplotlib.pyplot as plt



# Step 1: Generate a population

population_mean = 70

population_std = 15

population_size = 50000


population = np.random.normal(

    population_mean,

    population_std,

    population_size

)


# Step 2: Random sampling

sample_sizes = [20, 40, 80]

num_samples = 500


sample_means = {}


for size in sample_sizes:

    sample_means[size] = []
```

```
    for _ in range(num_samples):
        sample = np.random.choice(
            population,
            size=size,
            replace=False
        )

        sample_means[size].append(np.mean(sample))

# Step 3: Plot sampling distributions
plt.figure(figsize=(12, 8))

for i, size in enumerate(sample_sizes):
    plt.subplot(len(sample_sizes), 1, i + 1)

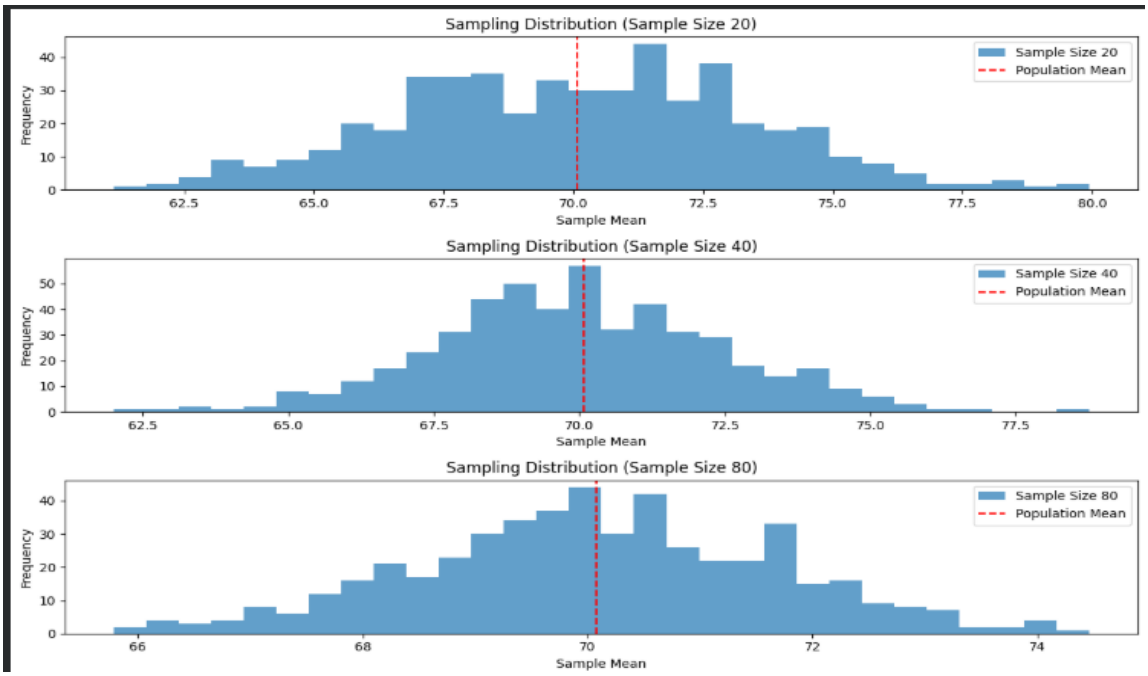
    plt.hist(
        sample_means[size],
        bins=30,
        alpha=0.7,
        label=f"Sample Size {size}"
    )

    plt.axvline(
        np.mean(population),
        color="red",
        linestyle="dashed",
        linewidth=1.5,
        label="Population Mean"
    )
```

```
    plt.title(f"Sampling Distribution (Sample Size {size})")
    plt.xlabel("Sample Mean")
    plt.ylabel("Frequency")
    plt.legend()

plt.tight_layout()
plt.show()
```

Output :



Result :

Thus, explore random sampling from a population and understand the concept of sampling distribution using Python in Jupyter Notebook.